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MMP+

GS 3 MAINS MODULE

UPSC CSE MAINS 2027

SCIENCE AND TECHNOLOGY



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Achievements of Indians in science & technology	
Name	**Key Contributions**
Jagdish Chandra Bose	• **Plant Physiology**: Demonstrated that plants respond to stimuli with electrical and physiological signals. • **Crescograph**: Invented a device to measure plant growth and movement precisely. • **Radio and Wireless Communication**: Conducted pioneering experiments with millimeter waves. • **Microwave Research**: Demonstrated microwave propagation and its similarity to light.
Prafulla Chandra Ray	• **Chemical Research**: Studied chemical reactions and compound synthesis. • **Father of Indian Pharmaceutical Industry**: Founded Bengal Chemicals. • **Education and Science Communication**: Promoted science through labs and journals.
C.V. Raman	• **Raman Effect**: Discovered in 1928; earned Nobel Prize. • **Raman Spectroscopy**: Developed for molecular analysis. • **Optics and Acoustics**: Studied light behavior and sound in musical instruments.
Meghnad Saha	• **Saha Ionization Equation**: Explained ionization in stellar gases. • **Saha Equation and Stellar Spectra**: Enabled spectral analysis of stars. • **Institute of Nuclear Physics**: Established in Kolkata for astrophysics research.
S.N. Bose	• Co-developed Bose-Einstein statistics • Laid theoretical foundation for quantum mechanics and Bose-Einstein Condensate.
Homi J. Bhabha	• Father of India's nuclear program • Proposed Bhabha scattering • Established TIFR and the Atomic Energy Commission.
Har Gobind Khorana	• **Decoding the Genetic Code**: Linked DNA to amino acid sequences. • **Synthesis of Nucleic Acids**: Created synthetic RNA/DNA for lab studies. • **Contributions to Genetic Engineering**: Enabled gene manipulation. • **Nobel Prize**: Won in 1968 for genetic code research
Vikram Sarabhai	• **Establishment of ISRO**: Founded India's space agency in 1969. **Initiating India's Space Program**: Focused on developmental applications. • **Development of Satellite Technology**: Led Aryabhata satellite launch. • **Promotion of Scientific Education and Research**: Indian Institute of Management Ahmedabad (IIMA), the Physical Research Laboratory (PRL)
S. Chandrasekhar	• **Chandrasekhar Limit**: Defined mass limit for white dwarfs. • **Stellar Structure and Evolution**: Modeled life cycles of stars. • **Theory of White Dwarfs**: Explained dense stellar remnants. • **Accretion Theory**: Described matter accumulation onto stars.

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M. Visvesvaraya	• **Design and Implementation of Irrigation Projects**: Designed Krishna Raja Sagara Dam. • **Flood Control and Water Management**: Innovated sluice gate systems. • **Reforms in PWD**: Improved infrastructure governance. • **Education and Institution Building**: Founded UVCE, promoted engineering education.
M.S. Swaminathan	• **Green Revolution in India**: Transformed agriculture during the 1960s. • **Development of High-Yielding Varieties**: Bred resilient rice and wheat. • **Crop Improvement through Hybridization**: Produced high-yield hybrids. • **Genetic Resource Conservation**: Advocated for seed and gene banks
P.C. Mahalanobis	• **Mahalanobis Distance**: Created statistical metric for data analysis. • **National Sample Survey**: Established data collection institution. • **Planning and Economic Development**: Designed Second Five-Year Plan. • **Foundation of ISI**: Founded Indian Statistical Institute. • **Agricultural Statistics**: Applied stats for rural planning. • **Sampling and Design of Experiments**: Innovated complex survey techniques.
Dilip Mahalanabis	• **Oral Rehydration Therapy**: Validated ORS in 1971 cholera epidemic. • **Cholera Control**: Worked with WHO globally. • **International Collaboration**: Contributed to bacterial disease control
A.P.J. Abdul Kalam	• **SLV-III Project and Rohini Launch**: Directed India's first satellite mission. • **Project Devil and Valiant**: Initiated missile R&D programs. • **Agni and Prithvi Missiles**: Developed India's strategic missile force. • **Pokhran-II Nuclear Tests**: Led 1998 tests as Chief Scientific Adviser. • **Kalam-Raju Stent**: Co-developed affordable heart stents. • **Kalam-Raju Tablet**: Created rural healthcare tech. • **Cryogenic Engine Development**: Pushed GSLV-D5 success. • **Leadership at DRDO/ISRO**: Advanced strategic defense systems. • **UAV Technology**: Led development of Lakshya, Nishant
C.N.R. Rao	• **Transition Metal Oxides**: Explored structure-property relations.. • **High-Temperature Superconductors**: Contributed to first liquid-N_2 superconductors. • **Y Junction Nanotubes**: Innovated carbon nanostructures.
S. Ramanujan	• **Theory of Partitions**: Advanced combinatorics and number theory. • **Mock Theta Functions**: Extended modular function theory. • **Black Hole Entropy and Quantum Physics**: Influenced string theory and quantum entropy. • **Impact on Modern Math & Physics**: Provided tools for quantum studies. • **Legacy -** Ramanujan's birth anniversary recognised as National Mathematics Day (Dec 22).

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Applications of Biotechnology	
Social (Food, Nutrition & Health)	• Food security: high-yield GM staples, climate-smart millets, RNAi pest-resistant vegetables. • Nutritional security: bio-fortified Golden Rice, iron-rich pearl millet, Zn-fortified wheat, probiotic functional foods. • Red biotech: mRNA/DNA vaccines, low-cost thermostable vaccines, CRISPR diagnostics (SHERLOCK/DETECTR), bioprinted organs & skin grafts, microbiome-based therapeutics for mental health. • Human Genome Project & beyond: population genomics (Genome India) enabling precision medicine, ancestry debunking racism myths. • Public-health outreach: portable lab-on-chip devices for rural antenatal & TB screening.
Agriculture & Allied (Green / Blue / Digital)	• GM & gene-edited crops – Bt cotton, Bt brinjal, omega-3 rich mustard, nitrogen-fixing cereal prototypes. • CRISPR/Cas “climate smart” lines – drought/salinity-tolerant rice, heat-resistant wheat, flood-rescue ‘scuba’ rice. • Bio-fertilisers & inoculants: phosphate-solubilising bacteria, mycorrhizal consortia; enhance soil C-sequestration. • Precision agriculture: drone- & AI-driven phenotyping, IoT soil sensors, blockchain farm-to-fork traceability. • Blue biotech: transgenic shrimp (WSSV-resistant), algae bio-stimulants, sea-weed (agar-agar, carrageenan) value chains, gene-banking for fish germplasm. • Post-harvest tech: edible nano-coatings, bio-degradable antimicrobials extending shelf-life.
Political / Governance	• Food sovereignty: domestic GM oilseeds & pulses reduce import bill. • Soft power: ‘Vaccine Maitri’ & genomics capacity-sharing in Global South. • Regulatory leadership: genome-edited crop draft guidelines, bio-economy (Bio-E3) roadmap, digital sequence information (DSI) negotiations. • IPR & farmers’ rights: Plant Variety Protection balancing breeders & traditional knowledge. • Public-funded biobanks: equitable access frameworks; NDHM health-data governance.
Security (Bio-defence & Bio-security)	• Pathogen early-warning – meta-genomic sewage surveillance, handheld CRISPR tests at borders. • Rapid antigen-switch vaccine platforms for emerging threats. • Digital watermarks in engineered organisms to trace illicit use. • High-containment labs (BSL-3/4) & “One-Health” integration for zoonotic spill-over control. • Forces protection – antidotes to nerve agents through enzyme biocatalysts.

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Legal & Forensic	• Rapid DNA 'lab-in-a-van' for disaster-victim ID within 90 minutes. • Epigenetic age & smoker status markers for forensic profiling. • Wildlife forensics – DNA barcoding to curb ivory & pangolin trade. • Gene-doping detection in sports via qPCR of performance-enhancing edits.
Environmental & Biodiversity	• Bio-remediation – plastic-eating PETase enzymes, PFAS-degrading microbes, oil-zapper consortia; myco-remediation of heavy metals. • Carbon sequestration – engineered rhizobacteria, ocean micro-algae sinks, biochar-producing microbes. • eDNA & drone-based biodiversity surveys detect elusive/endangered species. • Cryo-gene banks & synthetic biology to revive threatened flora/fauna (white-rhino embryo).
Industry & Biopharma	• Recombinant proteins & mAbs – next-gen bispecifics, antibody-drug conjugates, affordable biosimilars. • Cell & gene therapies – CAR-T (NexCAR19), CRISPR-Cas9 cures for sickle-cell, ex-vivo HSC edits. • Synthetic biology manufacturing – microbial cell-factories for bioplastics, sustainable pigments, cultured meat. • Smart drug-delivery – lipid-nanoparticle mRNA, pH-sensitive hydrogels, microneedle patches. • Continuous bioprocessing & single-use systems cut costs, boost pandemic surge capacity.
Energy & Green Industry	• Bio-fuels – algae-derived SAF (sustainable aviation fuel), cellulosic ethanol, green hydrogen via engineered cyanobacteria. • Bio-based chemicals – biosynthetic rubber, bio-PDO for eco-polyesters. • Microbial fuel cells convert wastewater to electricity.
Climate & Disaster Resilience	• Stress-tolerant crops to stabilise yields under extreme weather. • Pro-biotic coral & mangrove biotech for coastal defence. • Biosensors for early locust/pest alerts; pathogen spill-over sentinels.

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NANOTECHNOLOGY	
Definition	• Nanotechnology is a multidisciplinary field that involves the **manipulation and control of matter at the nanoscale,** which is typically within the range of 1 to 100 nanometers. • The unique properties and behaviour of materials at the nanoscale allow for **novel applications and functionalities -** enhanced properties, more efficient devices and systems, and innovative solutions in areas such as electronics, medicine, energy, environmental science, and more.
Applications of Nano-technology	
Technological	
Smart Packaging	• **Hybrid polymers:** Nanomaterials can be used to develop intelligent packaging that **changes color when food spoils or releases antimicrobial agents to extend shelf life, reducing food waste**
Triboelectric Nano Generators (TENGs)	• **Principle of creation of electrostatic charges** through the contact and separation of different materials, resulting in the conversion of mechanical energy into electrical energy. • **Self-Powered Devices:** Enabling self-sustaining electronic devices, reducing the need for external power sources. • **Biomedical Applications:** Providing a reliable and sustainable power source for devices such as pacemakers.
Gold NanoParticles (GNPs)	• GNPs are **biocompatible, have high surface area, more stability, and are non-toxic**. GNPs are melted at much lower temperatures (300°C) than bulk gold (1064°C) • **Advantages of GNP**: Greater solar radiation absorbing ability; photovoltaic cells, unique optical properties; therapeutic imaging for better resolution.
Electronics	• **Quantum dots** are used in **OLED screens** to enhance display quality with vibrant colors and higher efficiency. • **Graphene** is utilized in **flexible touch screens**, providing durability and flexibility. • **Carbon nanotubes (CNT) have the potential to replace silicon-based chips**, leading to faster and more efficient processors. • **Nanowires offer increased conductivity and strength** compared to traditional materials. • **Nano optomechanical SRAM (Static RAM)** provides faster read/write times compared to conventional RAM, improving overall device performance.
Space	• **Carbon nanotubes used to reduce the weight of the spaceships** while retaining or increasing the structural strength
Industry	• **Nano-molecular structures** to make asphalt and concrete more robust to wear seepage • **Heat-resistant nanomaterials** to block ultraviolet and infrared radiation • **Sunscreens:** Titanium dioxide and zinc oxide nano particles help block UV radiations
Social	
Improved	• **Nano sensors can detect harmful bacteria or contaminants** in food, such as E. coli or

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Food Safety	Salmonella
Enhanced Nutrition	• **Nanoencapsulation** techniques and **Nanocarriers** can protect and deliver essential nutrients, such as vitamins or antioxidants, ensuring their stability and targeted delivery to improve the nutritional value of food products
Medical	
Nanomicelles	• Nanomicelles are **globe-like structures with a hydrophilic outer shell and a hydrophobic interior.** This dual property makes them a perfect carrier for delivering drug molecules. • Advantage: Low toxicity, ability to minimize drug degradation, ability to permeate tissues easily • Recently, Researchers have found that Nanomicelles can be used for **cancer treatment**
Antigen Antibody tests	• Eg, **ELISA test** - A biochemical test that detects and quantifies specific substances, such as proteins, antibodies, or hormones, in biological samples using antibodies and enzymatic reactions.
Prosthetic Devices	• To improve strength, flexibility, functionality, and biocompatibility using **nanocomposite materials**.
Other Medical Uses	• **Nanohydrogel-based eye drops** deliver medications efficiently for eye conditions. • **Nano sensors and NanoFlares detect HIV and cancer** early with high accuracy. • **Gold nanoparticles enhance medical imaging** for clearer diagnostic images. • **Nanoscale devices** improve remote patient monitoring for continuous health tracking. • **Nanobots clear artery blockages**, providing a minimally invasive treatment for cardiovascular diseases. • **NanoSponges absorb toxins from the bloodstream**, treating poisoning and infections effectively. • **Gene sequencing technologies** through design and engineering of advanced solid-state **nano-pore materials.** ○ Eg IISER Bhopal conducted genome sequencing of Jamun using nanopore.
Political	
Reduce Import Dependency	• Nanotechnology-based solutions can help **increase agricultural productivity, develop drought-resistant crops,** and enhance post-harvest storage methods, contributing to food security and reducing reliance on food imports
Defence	• Nanotechnology enhances **tank polymers and camouflage** for **better stealth and durability.** • **Lightweight nanomaterials** improve the maneuverability of military vehicles. • **Precision-guided missiles** use nanotechnology for increased accuracy and performance.
Energy Security	• Nano-bioengineering of enzymes is applied in **biofuel production** to enhance efficiency. • **Solar panels** utilize advanced nanotechnology for improved performance. • **Nanotube scrubbers and electrostatic precipitators** increase efficiency in various

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	industrial processes. • **Gold nanoparticles (GNP)** are integrated into solar panels to boost energy capture. • **Carbon nanotubes (CNT)** are used in photovoltaic (PV) cells for better energy conversion. • **Graphene** is employed in the production of green hydrogen, contributing to sustainable energy solutions.
Legal	
Regulation and Safety Research	• Specific regulations can be established to ensure the safe use and disposal of nanomaterials in agriculture, addressing potential environmental and health risks associated with their production, application, and waste management
Agricultural	
Crop Protection	• **Nano pesticides** with controlled release mechanisms can target pests and diseases, reducing the amount of chemical inputs needed while minimizing their impact on the environment
Precision Agriculture	• **Nano sensors embedded in soil** can monitor moisture levels, nutrient content, and pest infestation, allowing farmers to precisely manage irrigation, fertilization, and pest control for optimal crop growth
Crop Enhancement	• **Nano fertilizers** can deliver nutrients directly to plant roots, improving nutrient absorption and reducing fertilizer runoff, leading to increased crop yields and decreased environmental pollution • **NUL (Nano Urea Liquid)** contains nano-scale nitrogen particles which have more surface area and number of particles making it more impactful ○ Reduces the requirement of conventional Urea **by 50%** or more and is cheaper than conventional urea • **Nanophosphorous fertilizer** is used to enhance the growth of millet and cluster beans. • **Silver nanoparticles** are effective against phytopathogens, helping to protect crops from diseases. • **Carbon nanoparticles** can be utilized to promote **seed germination.** • **Nickel/ferrite nanoparticles** serve as **antifungal agents,** combating fungal infections in plants. • Zinc oxide and chitosan nanoparticles are employed to **treat viral diseases** in crops. • **Nanoemulsions** are used to **reduce bacterial contamination** on farm produce, thereby minimizing post-harvest losses.
Environmental Technology	• **Goldene discovery**: world's thinnest gold leaf, measuring a mere single atom in thickness. Promising uses of goldene • **Carbon Dioxide Conversion**: Enhanced chemical reactivity of goldene can be used to convert CO2 into useful fuels like ethanol or methane. • **Climate Change Mitigation**: Potential use in converting carbon dioxide into useful fuels. • **Energy Production**: Revolutionizing hydrogen generation through water splitting

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Challenges related to Nanotechnology	
Social	• Nanotechnology may **exacerbate societal inequalities** if access to its benefits is not equitable.
Agriculture	• **Phytotoxicity and reactivity** of nanomaterials in the environment • **Cytotoxic and genotoxic effects** of cellular nanomaterials on Nano Agri Products • **High aspect ratio, stiffness and bio durability** of nano cellulose • Insufficient economic interest, regulatory issues and public opinion • **Lack of knowledge and developmental methods** for risk and life cycle assessment
Health	• **Triggers an acute inflammatory reaction** when enters in circulation system • **Biomagnification** in the food chain of nano particles
Environmental	• **Environmental contamination** due to nanoparticle release. • **High complexity** in assessing and managing nanotechnology impacts. • Fears of **self-replicating nanomaterials** causing uncontrolled proliferation.
Legal	• Policy framework and regulation development ○ Lack of risk assessment methodologies • Absence of a **unified regulatory body** to oversee its management.
Security	• Nano particles can be used in devices to **capture videos or use as drones** to launch offensive attack • Use of **Smart Matter/Smart Dust to spy:** Use of tiny wireless sensors and transponders to communicate the information and also have GPS capabilities
R&D	• **Equity imbalance**: Developing countries have less access to funds to create nanotechnology products • **Insufficient studies on toxicological studies** of nanoparticles • **Only 0.2% of the patents** filed in the US Patent Office are from India in nanotech • IPR protection can be difficult due to the interdisciplinary nature of the field and overlapping patents
Ethical	• Using nanotechnology and devices in human beings has potential to **create a more powerful human being** with enhanced human capabilities
International	• Collaboration among nations and the establishment of international standards are essential for harmonizing regulations, sharing knowledge, and promoting responsible development and use
Manufacturing	• Upscaling nanotechnology processes from the lab to large-scale production can be challenging, requiring advanced manufacturing techniques and quality control measures

Indian Initiatives in Nano-technology		
Infrastructure and Capacity	**Nano Science and Technology**	• Initiated in 2001 by DST for infrastructure development, research, and application programmes related to nano-materials including

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Building	**Initiative (NSTI)**	drugs, drug delivery, gene targeting, and DNA chips.
	Nano Mission	• Launched by Dr. CNR Rao in 2007 as an umbrella capacity-building programme.
	Nanotechnology Institutes	• Indian Institute of Nano Sciences in Bangalore, Mumbai, Kolkata.
	Center of Excellence in Nanoelectronics	• IIT Bombay and IISc Bangalore.
Research and Development Initiatives	**Nanotechnology Initiatives Programme**	• Started in 2004 by the Dept of Information Technology (DIT), Ministry of Communications and Information Technology (MCIT) focusing on nano-electronics aspects of nanotechnology.
	Triboelectric NanoGenerators	• Scientists from Centre for Nano and Soft Matter Sciences (CNSMS) have fabricated a cost-effective, bio-compatible nanogenerator.
	Synthesis of Gold Nanoparticles (GNPs) Using Psychrotolerant Antarctic Bacteria	• The National Centre for Polar and Ocean Research and Goa University successfully synthesized GNPs using psychrotolerant Antarctic bacteria.
	Decontamination of Arsenic from Water	• A team from IIT Madras used nanotechnology to decontaminate arsenic from water.
	Self-cleaning Technology for Textile Industry	• A team from IIT Delhi engineered a self-cleaning technology for use in the textile industry.
Regulatory and Safety Standards	**Nanotechnology Regulatory Board**	• Established in February 2010 to regulate industrial nanotechnology products used in daily life.
	Guidelines for Evaluation of Nano-based Agri-input and Food Products	• Launched in 2020 to help researchers develop products for agriculture and human consumption and regulators to assess the quality and safety of nano-based agriculture and food products.
	National Regulatory Framework Road-Map for Nanotechnology (NRFR-Nanotech)	• Developed under the Nano Mission.

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Commercial and Industrial Applications	**Nano Urea Liquid**	• National Fertilisers Limited (NFL) and Rashtriya Chemicals and Fertilisers (RCF) signed MoU with IFFCO for 'transfer of technology' of Nano Urea Liquid (NUL) fertilizer, making India the first country to start commercial production of NUL.
Human Resource Development	**Indian HR Development in Nanotechnology**	• Efforts led by the Nano Mission have made India one of the top five nations in scientific publications in nanoscience and technology (moving from 4th to the 3rd position). The Nano Mission has established national dialogues for R&D in nanotechnology standards.
Conferences and Collaborations	**ICONSAT**	• International Conference on NanoScience and NanoTechnology: biennial conferences under the aegis of the Nano Mission.

Way Forward to using Nano-technology (Life Cycle Approach)	
Production	
People	• Develop indigenous nanotechnology capabilities
Funds	• Increase investment in R&D
Infrastructure	• Public-private partnerships to facilitate technology transfer and commercialization of nanotechnology products
Regulation	• Develop a comprehensive regulatory framework specific to nanotechnology that ensures safety, environmental sustainability, and ethical considerations • Enhance monitoring and surveillance systems to track the use and impact of nanomaterials
IPR Development	
Patent Process	• Strengthening patent laws and streamlining the patent application process for nanotechnology innovations
Industry Academia Linkages	• Encourage collaboration between research institutions, industry, and legal experts to support the effective protection and commercialization of nano-related intellectual property
Leadership	
Collaboration and Partnership	• Participate actively in global forums and initiatives related to nanotechnology to share knowledge, exchange best practices, and leverage international expertise • Foster international collaboration and partnerships with leading countries • Collaborate with international organizations to establish common standards and guidelines for nanotechnology research, manufacturing, and safety

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Antimicrobial Resistance	
Definition	• Antimicrobial Resistance (AMR) occurs when microbes, such as bacteria, fungi, parasites, and viruses, evolve to the extent that antimicrobial drugs, which are designed to kill or inhibit them, become ineffective. This resistance develops due to the microbes' ability to change in response to the use of these drugs. The World Health Organization (WHO) identifies AMR as one of the top 10 global public health threats. • It is closely linked to the **triple planetary crisis** of climate change; biodiversity loss; and pollution and waste.
Causes of AMR	• **Overuse and Misuse of Antimicrobials** • **Poor Infection Control in Healthcare:** Inadequate sanitation and hygiene in hospitals • **Environmental Discharge:** Release of untreated waste from hospitals and industries contributes to AMR.
Implications of AMR	• **Health Impact:** Increases in mortality, morbidity, and prolonged illness (1.27 million deaths in 2019). • **Economic Impact:** Longer hospital stays, additional tests, and potential GDP losses of USD 1 trillion to 3.4 trillion annually by 2030 (World Bank).
Initiatives to Combat AMR	
National	• **National Program on AMR Containment:** Aims to reduce AMR through improved practices and responsible use. • **National Action Plan on AMR (NAP-AMR), 2017:** Focuses on a One Health approach and involves multiple stakeholders. • **National AMR Surveillance Network (NARS-Net):** Collects and analyzes AMR data across India. • **Drugs and Cosmetics Rules, 1945:** Regulates antimicrobial use to prevent misuse.
Global	• **Global Action Plan on AMR (WHO):** Provides a framework for tackling AMR. • WHO's Core Package Interventions on AMR. • **Global Antimicrobial Resistance and Use Surveillance System (GLASS), 2015.** • **Global Antibiotic Research and Development Partnership (GARDP):** A WHO-backed nonprofit.

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	• **SECURE Program:** Developed by GARDP and WHO to expand access to essential antibiotics in low and middle-income countries (LMICs). • **Carb-X:** A global nonprofit partnership aimed at accelerating antibacterial innovation.
Way Forward to Combat AMR	• **Defined Antibiotic Use Policy in Medical Institutes** • **Promote Use of Alternatives to Antibiotics** • **Expansion of Existing Financing Instruments** • **Update the 2015 Global Action Plan:** Quadripartite organizations (FAO, UNEP, WHO, WOAH) should revise and update the Global Action Plan to reflect new challenges and strategies for combating AMR effectively.

Fixed-Dose Combinations (FDCs) Drugs	
Context	**The Ministry of Health and Family Welfare prohibited the manufacture, sale or distribution of 156 fixed-dose combination (FDC) medicines.**
What are Fixed-Dose Combinations (FDCs)?	• FDCs are pharmaceutical products that contain two or more active ingredients combined in a specific ratio, used to treat particular medical conditions **(as defined by the Drugs & Cosmetics Rules, 1945).** • **An FDC is considered a new drug if it meets either of the following conditions:** ○ It combines two or more drugs, previously approved for individual use, in a fixed ratio. ○ It alters the ratio of drugs in an already approved FDC and changes its claims.
Rationale for Using FDCs	• **Enhanced Efficacy** • **Cost-Effectiveness** • **Reduced Pill Burden** • **Improved Adherence**.
Issues Associated with FDCs	• **Lack of Dose Flexibility** • **Unapproved and Banned FDCs:** Easy access to untested and unlicensed FDCs. • **Increased Risk of Antimicrobial Resistance (AMR)** • **Limited Patient Choice**
Steps taken	• **Drugs & Cosmetics (Amendment) Act, 2008:** Harsher penalties for spurious drugs; certain offences made cognizable and non-bailable.

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	• **CDSCO Labs:** Testing capacity being upgraded for faster drug analysis. • **Drugs and Cosmetics Rules 1945 amended in 2017:** Bioequivalence study mandatory for oral drug license applications.
Way Forward	• Ensure robust scientific evidence for the efficacy and safety of FDCs to avoid unjustifiable combinations. • Implement strict post-market surveillance systems to quickly detect and address adverse effects related to FDCs. • Align export policies with domestic regulations to prevent the export of FDCs that are banned or restricted in India. • The CDSCO should enforce stringent guidelines for the approval of FDCs.

Chimeric Antigen Receptor-T-cell therapy	
Definition	CAR-T therapy is a type of immunotherapy where a patient's own T-cells (a type of white blood cell) are genetically modified to better recognize and destroy cancer cells.
Principle (5-step cycle)	1. **Leukapheresis**: patient's T-cells collected 2. **Gene insertion**: lab adds a synthetic receptor (CAR) that locks on to a tumour antigen 3. **Cell expansion**: modified T-cells multiplied to millions 4. **Infusion**: cells returned after short lymphodepleting chemo 5. **Attack & persistence**: CAR-T cells hunt and kill cancer cells
India milestones	• **NexCAR19** (ImmunoACT + IIT-Bombay) • **Var-cel** – second indigenous CAR-T cleared **Jan 2025** • **National CAR-T Programme** (DBT/CDSCO) scaling GMP facilities and training
Benefits	• Single-shot, highly personalised • 60–90 % remission in otherwise untreatable blood cancers
Key risks	• Cytokine Release Syndrome • Neuro-toxicity • Antigen escape • High upfront cost
Next frontiers	• "Off-the-shelf" allogeneic CAR-T • Multi-antigen & armoured CAR-Ts for solid tumours • Insurance/Ayushman Bharat coverage to widen access

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Gaganyaan: India's Indigenous Human Spaceflight Programme

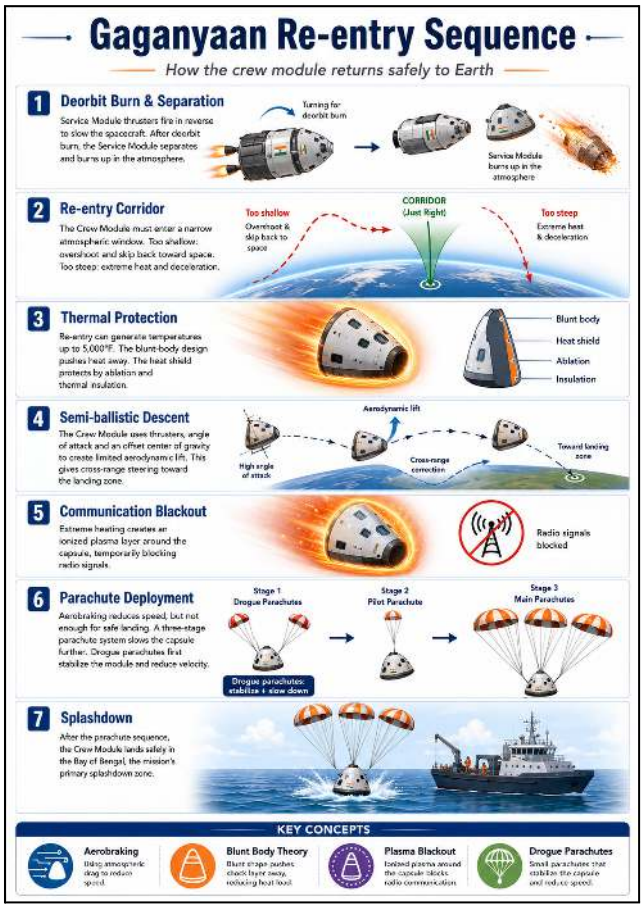
Gaganyaan is India's first indigenous human space mission, representing a culmination of a series of uncrewed and crewed flights designed to establish India's capacity for human spaceflight. It acts as a critical stepping stone for India's long-term space roadmap, which includes building the **Bharatiya Antariksh Station (BAS)** by 2035 and achieving an **Indian human landing on the Moon by 2040.**

Mission Architecture and Components

- **Launch Vehicle:** ISRO is utilizing the Human-Rated Launch Vehicle Mark-3 (HRLV Mk-3) for the mission. The launch vehicle comprises S200 motors, an L110 liquid stage, and a C32 cryogenic stage.
- **Orbital Module:** This is the spacecraft that will house the astronauts and provide them with life support, consisting of two main parts:
    - **Crew Module (CM):** The habitable module where astronauts will reside. It is currently undergoing thermo-vacuum tests at the U.R. Rao Satellite Centre prior to its final integration.
    - **Service Module (SM):** The unpressurized module providing propulsion and power.
- **Crew Escape System:** A critical safety element integrated into the vehicle to pull the crew module away from the rocket in case of an abort.

Phasing of the Mission

- To ensure the utmost safety of the astronauts, ISRO is executing a phased approach with three uncrewed test missions (Gaganyaan-G1, G2, and G3) before the final manned mission.
- **Gaganyaan-G1:** The maiden uncrewed test mission aims to simulate actual human spaceflight conditions. It will place the orbital module into an initial 170 km x 430 km elliptical orbit, which will subsequently be circularized.



Re-entry and Safe Return (The Most Critical Phase)

Safely returning astronauts from space requires dissipating immense kinetic energy (over 98% as heat) and drastically reducing orbital velocity. ISRO has previously tested relevant technologies via the Space Capsule Recovery Experiment (SRE) in 2007 and the Crew Module Atmospheric Re-entry Experiment (CARE) in 2014.

The Gaganyaan re-entry sequence involves:

- **Deorbit Burn and Separation:** The Service Module (SM) thrusters perform a reverse thrust (a 180-degree turn) to reduce speed. After this, the SM separates from the Crew Module and burns up in the atmosphere.
- **The Re-entry Corridor:** The Crew Module must enter a narrow atmospheric window. If the angle is too shallow (overshoot boundary), the capsule could bounce back into space; if it is too steep (undershoot boundary), the extreme heat and deceleration would be fatal.
- **Thermal Protection:** To survive surface temperatures of up to 5,000°F (as seen in deep space missions), the capsule uses the "blunt body theory," where a rounded forebody deflects heat away. The heat shield protects the capsule via ablation (where material burns away to carry off heat) or thermal insulation (using low-conductivity materials).
- **Semi-Ballistic Descent:** The Crew Module operates as a semi-ballistic body using its thrusters. By flying at a specific angle of attack and offsetting its center of gravity, it forces air to flow asymmetrically, creating

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aerodynamic lift. This provides the capsule with a cross-range capability to steer and glide toward its landing zone.

- **Communication Blackout:** The extreme heat of re-entry creates an ionized plasma layer around the capsule, temporarily blocking radio signals.
- **Splashdown:** While aerodynamic drag (aerobraking) slows the capsule significantly, its terminal velocity is still too high for a safe landing. A three-stage parachute system is deployed to further reduce speed, ensuring a safe splashdown in the Bay of Bengal, which is the primary landing zone for the mission.
    - **Drogue parachute tests for Gaganyaan mission successful:** Drogue parachutes play a pivotal role in stabilising the crew module and reducing its velocity to a safe level during re-entry

Significance as per ISRO:

The Human spaceflight programme has both **tangible and intangible benefits** for the nation, which include:

- Progress towards a sustained and affordable human and robotic programme to explore the solar system and beyond.
- Advanced technology capability for undertaking human space exploration, sample return missions, and scientific exploration.
- Future capability to actively collaborate in global space station development & to carry out scientific experiments of interest to the nation.
- Create a broad framework for wider academia-industry partnerships in taking up development activities for national development.
- Ample scope for employment generation and human resource development in advanced science and R&D activities.
- Unique opportunity to inspire and excite Indian youth and steer many students toward careers in science and technology, towards challenging jobs that encourage knowledge, innovation, and creativity.
- The programme will strengthen international partnerships and global security through the sharing of challenging and peaceful goals. Having a vibrant human spaceflight programme can be leveraged as a potent foreign policy tool.

NISAR

- **National Aeronautics and Space Administration - Indian Space Research Organisation Synthetic Aperture Radar (NISAR)** is the first major joint Earth-observation satellite mission of **National Aeronautics and Space Administration (NASA)** and **Indian Space Research Organisation** .
- It was launched on **30 July 2025** from **Satish Dhawan Space Centre, Sriharikota**, onboard **Geosynchronous Satellite Launch Vehicle-F16 (GSLV-F16)**. It is now in the **science phase**, after launch, deployment and commissioning.
- It uses **dual-frequency Synthetic Aperture Radar**, namely **L-band radar by NASA** and **S-band radar by ISRO**, enabling all-weather, day-and-night monitoring of Earth's surface.

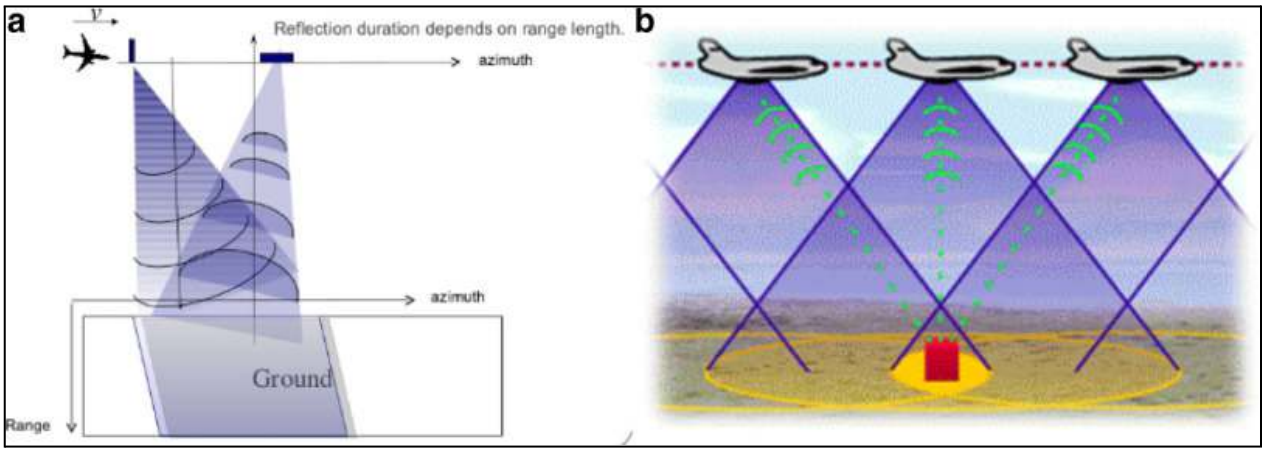
Working of NISAR

NISAR uses **radar signals** instead of **visible light** to monitor changes in land, water, vegetation, ice, and infrastructure. Its core value is not merely imaging the Earth, but detecting change: ground movement, soil moisture variation, forest biomass change, crop condition, glacier movement, shoreline change, and disaster damage.

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Synthetic Aperture Radar (SAR) is a technology that creates **high-resolution images** using **microwave pulses** instead of visible light, making it effective in **dark or cloudy conditions**. SAR systems emit **microwave signals** and record the **echoes** reflected from surfaces like **ground, ocean, ice, or buildings**.

- A **small antenna** mounted on a **moving satellite** captures echoes from different positions; **software stitches them together**, simulating a **large antenna** and enhancing image resolution.
- **Microwaves can penetrate clouds, smoke, light rain**, enabling SAR to operate **day & night in all weather conditions**. SAR satellites can **map wide swaths of land** in a single pass and **differentiate materials** (soil, vegetation, water, metal) based on how they reflect microwaves.

Applications

1. Disaster Risk Reduction:

- **Floods:** Real-time mapping of inundation, recession, waterlogging and damage during cloudy monsoon conditions.
- **Earthquakes and landslides:** Detection of ground deformation, unstable slopes and post-disaster surface changes.
- **Cyclones and coasts:** Monitoring of shoreline change, storm surge impact and coastal erosion.

2. Climate Change and Ecosystem Monitoring:

- Tracks glaciers, ice sheets, snow cover and Himalayan ground instability.
- Monitors mangroves, wetlands, shoreline retreat and coastal land loss.

3. Agriculture and Food Security:

- Tracks crop growth, crop stress, cropping intensity and flood/drought impact.
- Supports soil moisture assessment, irrigation planning and drought preparedness.
- Works effectively during cloudy monsoon periods, unlike optical satellites.

4. Forests, Biodiversity and Carbon Sink:

- Tracks forest degradation, fragmentation, wetland change and mangrove health.
- Strengthens monitoring of carbon sinks and ecosystem restoration.

5. Water, Land and Urban Governance:

- Maps surface water, reservoirs, floodplains and seasonal water spread.
- Supports monitoring of land-use change, urban expansion and ecological degradation.

6. Strategic and Scientific Value

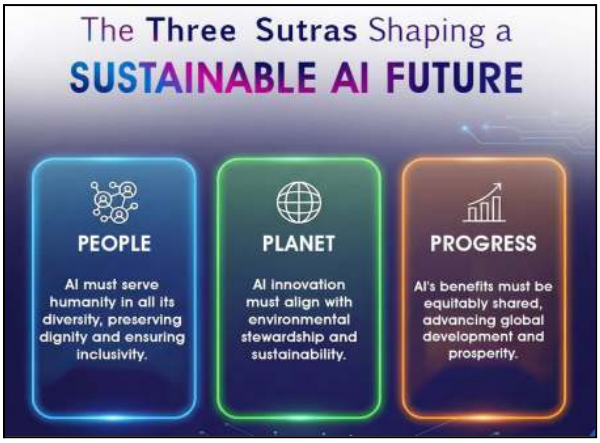
- Reflects **India-United States technology cooperation** in advanced Earth observation.
- Strengthens India's **remote-sensing ecosystem** through ISRO's S-band radar, spacecraft bus and operations role.

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India AI Impact Summit 2026



The India - AI Impact Summit is anchored on **three foundational pillars, known as 'Sutras'.** These three Sutras define how AI can be harnessed through multilateral cooperation for collective benefit.

Themes for Global Cooperation on AI

The **Seven Chakras** translate the guiding Sutras into concrete areas of multilateral action. These themes, spanning **human capital, inclusion, trust, resilience, science, resources, and social good**, channel global collaboration towards measurable outcomes.

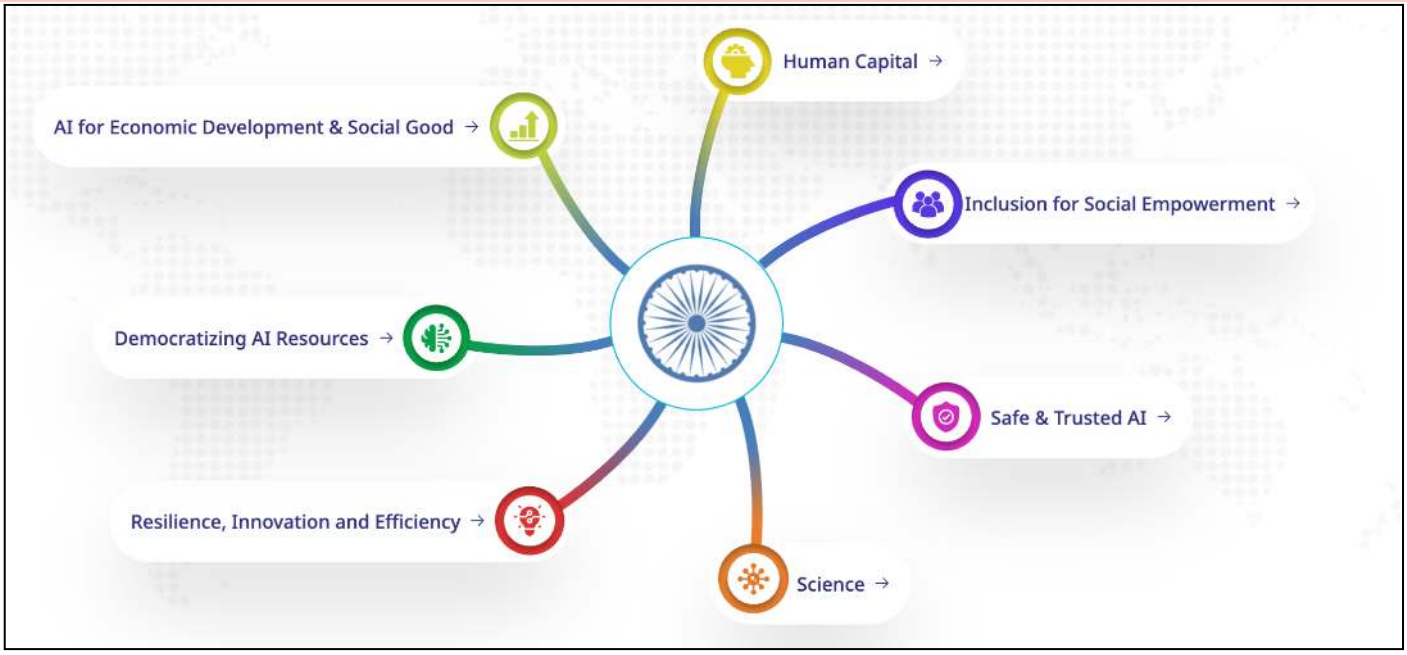
New Delhi Declaration

- The AI Impact Summit 2026, held in New Delhi on 18-19 February, concluded with the adoption of the **New Delhi Declaration on AI Impact**, marking a significant milestone in global cooperation on artificial intelligence.
- The Declaration has been **endorsed by 92 countries and international organisations**, reflecting a broad-based global consensus on leveraging AI for economic growth and social good.

A Shared Global Vision for AI

- Guided by the principle of ***"Sarvajan Hitaya, Sarvajan Sukhaya"*** (Welfare for all, Happiness for all), the Declaration underscores that the benefits of AI must be **equitably shared across humanity**.
- It emphasizes:
  - Strengthening **international cooperation and multistakeholder engagement**
  - Respecting **national sovereignty**
  - Advancing AI through **accessible, and trustworthy frameworks**

Seven Pillars (Chakras) of Action



Chakra	Focus Areas	Indian Significance	Global Significance
Human Capital	Advancing equitable skilling and inclusive workforce	India has a young population, but the demographic dividend can become a demographic burden if	Countries that gain most from AI will not be those that merely import AI tools, but those that

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	transitions for an AI-enabled future of work. **AI Workforce Development Playbook & Reskilling Principles** • Supports AI skilling, reskilling, and literacy • Prepares nations for an AI-driven economy	workers are not prepared for AI-led changes.	train their workforce to use, adapt and build them.
Inclusion for Social Empowerment	Advancing AI systems that are inclusive by design, empowering diverse communities and ensuring equitable representation. **AI for Social Empowerment Platform** • Enables knowledge exchange and scalable solutions • Focuses on equitable AI adoption	AI can be layered onto platforms for language translation, voice-based governance, disability support, rural health advice and education.	If AI is multilingual, accessible and designed for public use, it can democratise education, healthcare, governance and livelihood opportunities.
Safe and Trusted AI	Building globally trusted AI systems anchored in transparency, accountability, and shared safeguards for innovation. **Trusted AI Commons** • Repository of tools, benchmarks, and best practices • Supports development of secure and trustworthy AI systems	For India, trusted AI is especially important because of scale and diversity. A biased AI system can affect millions if used in welfare delivery, policing, recruitment, credit, education or healthcare.	Going ahead, this Chakra means countries may increasingly cooperate on common testing standards, safety benchmarks, watermarking, AI audit methods and risk-management practices.
Science	Harnessing AI to accelerate frontier science, foster scientific collaboration, and translate breakthroughs into shared global progress. **International Network of AI for Science Institutions** • Facilitates global scientific collaboration • Enhances AI-driven research capabilities	For India, AI for science matters in areas like weather forecasting, monsoon modelling, drug discovery, genomics, space research, agricultural research, materials for clean energy and public health.	AI is becoming a "general-purpose scientific instrument." Just as the microscope changed biology and the telescope changed astronomy, AI can change the pace of discovery across disciplines.
Resilience, Innovation, and Efficiency	Driving sustainable, resource-efficient AI systems that strengthen climate resilience and sustainability. **Guiding Principles on Resilient & Efficient AI**	India must balance AI expansion with energy security and climate commitments	future AI race cannot be only about bigger models. It also has to be about cheaper, smaller, greener and more efficient models.

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	• Focus on energy-efficient AI systems • Supported by a Playbook on AI Infrastructure Resilience		
Democratizing AI Resources	Promoting equitable access to foundational AI resources for inclusive innovation and sustainable development worldwide. **Charter for the Democratic Diffusion of AI** • Promotes affordable access to foundational AI resources • Supports locally relevant innovation ecosystems	For India, this directly connects with **IndiaAI Mission**, sovereign compute capacity, Bhashini, open datasets, Digital Public Infrastructure and startup-led AI innovation.	For the world going ahead, this Chakra means AI governance will not only be about controlling risks; it will also be about **fair access to AI infrastructure**. It brings the idea of an **AI digital public good** into global diplomacy.
AI for Economic Development & Social Good	Leveraging AI to enhance productivity, innovation, and inclusive development across economies and societies. **Global AI Impact Commons** • A platform to scale and replicate AI use cases globally • Enables cross-country collaboration for development impact	For India, this matches the developmental use of technology: Aadhaar, UPI, CoWIN, ONDC and DigiLocker showed how digital platforms can work at population scale. AI can be the next layer over this public digital infrastructure.	This Chakra could create a global library of proven AI use cases that countries can adapt instead of starting from zero.

Previous Global AI Summits

AI Summit	Key Outcome
AI Safety Summit, Bletchley Park, UK (2023)	• The first global summit centred on **frontier AI safety**, especially the risks of deliberate misuse and loss of human control. • The **Bletchley Declaration** established the need for shared understanding, testing and international cooperation on advanced AI risks. • AI companies agreed to provide governments early access to powerful models for safety evaluation, while the UK's Frontier AI Taskforce became the permanent **UK AI Safety Institute**
AI Seoul Summit, South Korea (2024), Co-hosted with the UK	• The summit translated the Bletchley discussion into more practical safeguards. Sixteen leading technology companies adopted voluntary **Frontier AI Safety Commitments** for responsible development of advanced models. • Governments agreed to create an international network of **AI Safety Institutes** and the agenda widened to include **safety, innovation and inclusion**

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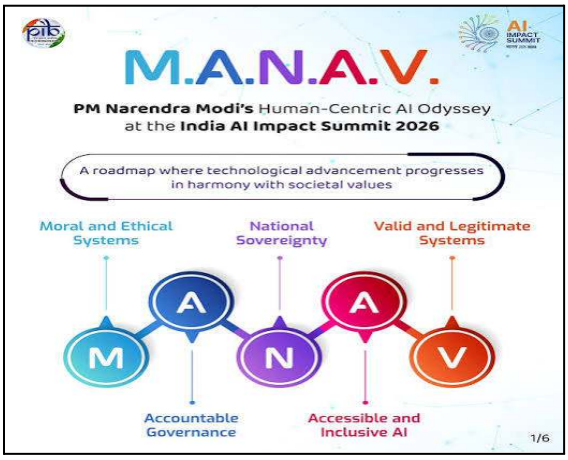
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AI Action Summit, Paris, France (2025)	• The focus shifted from safety to **innovation, economic opportunity, sustainability, employment, and equitable access**, particularly for the Global South. • The summit supported multilateral and multistakeholder governance, recognised the UN's **Global Digital Compact**. • Sixty countries backed a non-binding **Pledge for a Trustworthy AI in the World of Work**, while the United States and United Kingdom did not sign the final statement

M.A.N.A.V. Framework for Human-Centric AI

At the **India AI Impact Summit 2026**, Prime Minister Narendra Modi presented the **M.A.N.A.V. framework** as a human-centric roadmap for building AI that advances innovation while protecting ethics, accountability, sovereignty, inclusion and trust.



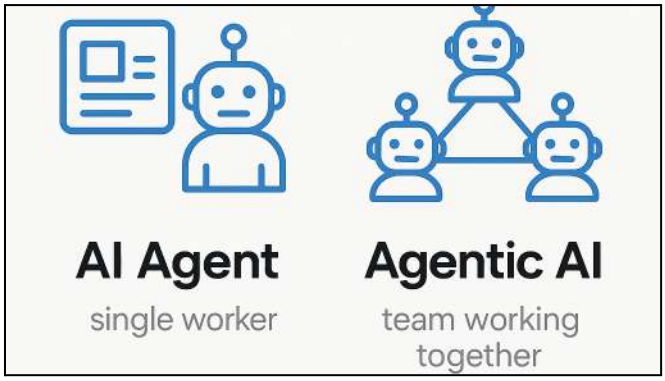
Pillar	What It Means
M - Moral and Ethical Systems	• AI must be guided by **fairness, transparency and human oversight**. The aim is to ensure that technology supports human dignity rather than replacing ethical judgment.
A - Accountable Governance	• AI systems need **clear rules, explainability and institutional oversight**. Initiatives such as the **IndiaAI Mission** and AI Governance Guidelines seek to make development and deployment more responsible and auditable.
N - National Sovereignty	• In the AI era, sovereignty also includes control over **data, algorithms, chips, cloud and digital infrastructure**. India seeks global cooperation while building secure domestic capabilities through initiatives such as the **India Semiconductor Mission**.
A - Accessible and Inclusive AI	• AI should act as a **multiplier, not a monopoly**. Shared digital infrastructure, compute platforms and datasets should allow startups, researchers and public institutions to use AI in healthcare, education, agriculture and governance.
V - Valid and Legitimate Systems	• AI must be **lawful, verifiable and safe**. This includes safeguards against deepfakes, bias, privacy violations and harmful synthetic content, supported by auditing and risk-assessment mechanisms.

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Agentic AI



- An **AI agent is a software program that autonomously perceives its environment, reasons, plans, and takes actions to achieve specific goals**, moving beyond simple text generation to complete multi-step tasks. Powered by Large Language Models (LLMs), **these agents can use external tools such as APIs, web browsers, or software to perform tasks like sending emails, coding, or managing projects with minimal human oversight.**
- **Agentic AI** refers to **Artificial Intelligence systems that can independently plan, decide, and perform tasks to achieve a goal with minimal human intervention**.
- Unlike traditional AI that only responds to direct prompts, **Agentic AI can act like an "agent" by taking initiative, using tools, and adapting based on results.**
    - **Example**: **AI coding assistants**: Read code, edit files, debug, **Customer service AI** to Resolve complaint workflow, **Supply-chain AI** to Manage inventory automatically, etc.

Basis	Agentic AI	Generative AI
Meaning	AI that can **plan, decide, and act autonomously** to achieve goals	AI that can **create new content** such as text, images, audio, video, or code
Main Function	Executes tasks and takes actions	Generates outputs based on prompts
Autonomy Level	High – can work with minimal supervision	Low to moderate – usually needs user prompts
Decision-Making	Makes step-by-step decisions during task execution	Usually responds only to given input
Human Role	Gives objective, AI handles workflow	Gives prompts repeatedly for outputs

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Anusandhan National Research Foundation (ANRF)

The **ANRF Act, 2023** created the Anusandhan National Research Foundation as India's apex institution for research, innovation and entrepreneurship. It replaces the earlier Science and Engineering Research Board and functions under the Department of Science and Technology. It aims to mobilise funds amounting to ₹50,000 crore during 2023–28 through multiple streams including the ANRF Fund, Innovation Fund, Science and Engineering Research Fund, and Special Purpose Funds.

Objectives of ANRF

- **Expand research across institutions:** Seed and strengthen research in universities, colleges, laboratories and other recognised institutions, especially those with weak research capacity.
- **Support high-impact science:** Fund basic, interdisciplinary and mission-oriented research linked with national priorities and emerging technologies.
- **Connect research with society and industry:** Encourage collaboration among academia, laboratories, start-ups, industry, philanthropy and government.
- **Improve research access:** Make funding, infrastructure and scientific knowledge more widely available, including through tools such as **SARAL**.

SARAL Initiative

- **Making research easier to understand:** SARAL- **Simplified and Automated Research Amplification and Learning**- uses Artificial Intelligence to convert complex research papers into simpler summaries for students, policymakers and the public.
- **Multiple formats and languages:** It can transform research into **videos, podcasts, posters, presentations and short-form content**, including output across multiple Indian languages.
- **Bridging science and society:** SARAL does not replace the original research paper; it helps wider audiences understand and use scientific findings that would otherwise remain limited to specialists.

Key Features

- **Apex research body:** The Governing Board is presided over by the Prime Minister, while the Executive Council is chaired by the Principal Scientific Adviser.
- **Broad research mandate:** ANRF supports natural sciences, engineering, health, agriculture, environmental sciences and the interface of science with humanities and social sciences.
- **Support across career stages:** Its programmes cover early-career researchers, postdoctoral fellows, established scientists and senior scientific leaders.
- **Focus on weaker institutions:** The **PAIR programme** connects leading institutions with State universities and other less-endowed institutions through a hub-and-spoke mentoring model.



- **Research-to-impact approach:** ANRF combines investigator-driven grants with mission programmes in areas such as electric mobility, MedTech, critical minerals, advanced materials and AI for science.

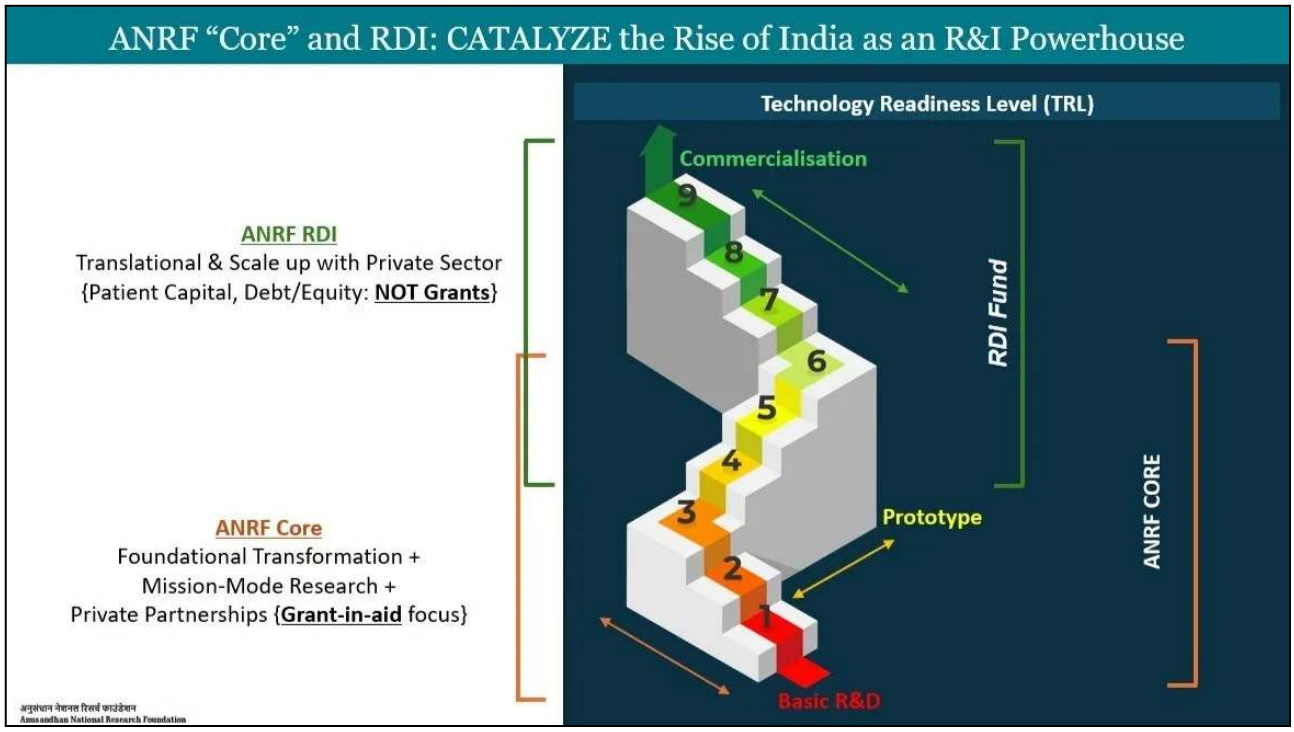
Research, Development and Innovation Fund

The **₹1 lakh crore Research, Development and Innovation Fund** is a Special Purpose Fund under ANRF. It complements ANRF's research grants by helping promising technologies move from laboratory development towards commercial scale.

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How it is linked with ANRF

- **ANRF Core builds the research pipeline:** ANRF grants support universities, laboratories and researchers working on basic and applied research, generally at lower Technology Readiness Levels.
- **RDIF supports translation and scaling:** Once a technology reaches **Technology Readiness Level 4 or above**, RDIF can support private companies and start-ups in developing prototypes, scaling production and reaching the market.
- **Complementary, not identical:** ANRF Core mainly provides **grants**, whereas RDIF provides **patient capital that is expected to be repaid or generate returns**.



Key Features of RDIF

- **₹1 lakh crore over six years:** It is designed to expand private investment in research-intensive and deep-technology sectors.
- **Not a grant scheme:** Support may be provided through **long-term loans, debt, equity or hybrid instruments**, rather than ordinary grants.
- **Patient capital:** Funding is intended to have longer repayment periods and lower costs so that firms can survive the high risks and long development cycles of advanced technologies.
- **Two-level funding structure:** ANRF acts as the first-level custodian, while **Second-Level Fund Managers**-such as Alternative Investment Funds, Development Finance Institutions, Non-Banking Financial Companies and Focused Research Organisations-select and finance individual projects.
- **Focus on strategic sectors:** Priority areas include **AI, quantum technology, robotics, space, biotechnology, medical devices, energy transition, climate technologies and the digital economy**.
- **Support for Indian control and intellectual property:** Eligible entities must be based in India, remain under resident Indian control and register RDIF-supported intellectual property in India.

Key Achievements of ANRF

- **Operational research architecture established:** ANRF has made its framework functional across foundational research, mission-mode programmes, industry-academia partnerships and the RDI pipeline.
- **Support to young researchers expanded:** The **Prime Minister Early Career Research Grant** has already supported **718 young scientists** through more flexible and research-friendly funding norms.
- **University research capacity strengthened:** Under **PAIR Category-A**, **45 institutions** have joined networks led by stronger research institutions, improving mentorship, proposal quality and research capability.

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- **Mission-mode research advanced:** Programmes in **MedTech, advanced materials, 2D materials and AI for science** have moved research towards clearly defined national and technological priorities.
- **Co-funding ecosystem widened:** ANRF has brought in partners such as the **Gates Foundation, Wadhwani Foundation and industry bodies**, helping shift Indian research from isolated public funding towards a **whole-of-government and whole-of-society model**.

Key Challenges

- **Government-heavy decision-making:** Limited representation of State universities, practising researchers, young scientists and Indian industry may weaken ground-level understanding.
- **Slow administrative processes:** Delays in approvals, procurement, customs clearance and fund release can make research equipment or technologies outdated before use.
- **Unequal institutional capacity:** Better-funded institutions are more likely to win grants, while weaker universities may struggle with proposals, laboratories and grant management.
- **Private funding uncertainty:** The model expects substantial non-government participation, but industry may prefer commercially safer projects over basic or long-term public-interest research.
- **Risk of fragmented implementation:** Multiple committees, funding channels and programme structures may recreate the bureaucracy ANRF was meant to reduce.

Way Forward

- **Broaden representation:** Include more State-university faculty, young researchers, entrepreneurs, women scientists and industry experts in decision-making bodies.
- **Fix clear timelines:** Set time limits for peer review, sanction, fund release and procurement, with public tracking of delays.
- **Strengthen weaker institutions:** Provide proposal-support cells, shared laboratories, research managers and sustained mentoring rather than only competitive grants.
- **Protect basic research:** Maintain a clear balance between commercially useful technologies and investigator-driven fundamental research.
- **Measure real outcomes:** Evaluate ANRF through research quality, prototypes, technology transfer, institutional capacity and societal impact, not only through funds sanctioned or papers published.

Case study: Defense Advanced Research Projects Agency (DARPA)

- The United States' **Defense Advanced Research Projects Agency (DARPA)** shows how a public research body can remain agile. It recruits expert programme managers from academia and industry for limited tenures and gives them the authority to fund **high-risk, high-reward projects** against clear technical milestones.
- This flexibility helped support breakthrough technologies such as the **internet, GPS and autonomous systems(Tesla's Autopilot)**, offering ANRF a model for faster decisions and stronger research-to-application pathways.

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India's Three-Stage Nuclear Power Programme

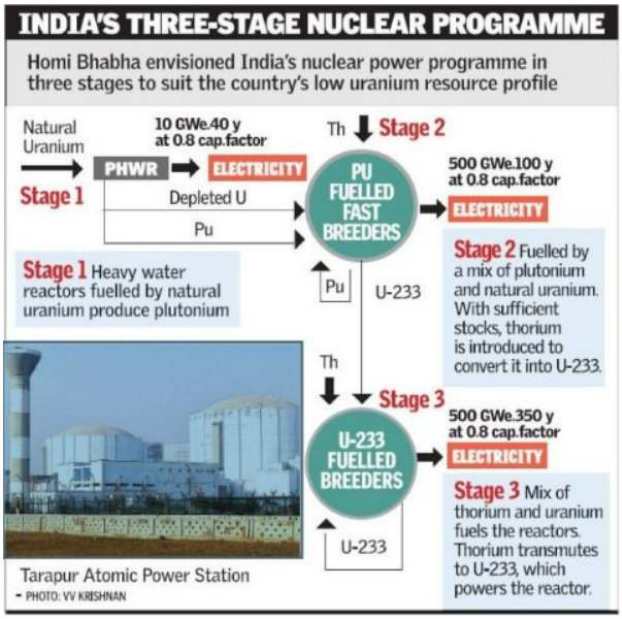
Current Nuclear Power Landscape in India

India's nuclear energy programme has maintained a steady presence in the country's electricity mix. It now stands at a pivotal moment, with significant expansion planned over the coming years.

- **Installed Capacity:** India's current nuclear capacity is **8.78 gigawatts (GW)**. In 2024–25, nuclear power plants generated 56,681 Million Units of electricity across the country.
- **Stable Contribution:** Nuclear power has consistently accounted for around 3% of India's total electricity generation. In 2024–25, its share stood at 3.1%.
- **Planned Expansion:** India's nuclear capacity is set to grow nearly 3x in the coming years. With indigenous 700 Megawatt (MW) reactors and 1,000 MW reactors being developed through international cooperation, the installed capacity is projected to reach **22.38 GW** by 2031–32.

The three-stage Programme

Dr. Homi J. Bhabha designed India's three-stage nuclear power programme to ensure **long-term energy security** by using India's **limited uranium reserves** and **large thorium reserves**. Its core idea is a **closed fuel cycle** in which the spent fuel from one stage becomes the input for the next.



Why India Chose This Model

- India has **limited domestic uranium** but **abundant thorium**; therefore, the programme aims to gradually shift from uranium dependence to thorium utilisation.
- The programme **links Pressurized Heavy Water Reactors (PHWRs), Fast Breeder Reactors (FBRs) and thorium reactors** to reduce long-term dependence on imported nuclear fuel and technology.
- India follows **"reprocess to reuse",** where spent fuel is treated as a resource rather than waste.

Stage-wise Working and Status

Stage	Reactor / Fuel	Working	Purpose and Status
Stage 1	**Pressurised Heavy Water Reactors (PHWRs)** using **natural uranium**	• PHWRs use natural uranium and heavy water. • During operation, Uranium-238 gets converted into Plutonium-239 in spent fuel. • The spent fuel is reprocessed to extract plutonium.	• Generate electricity and produce **Plutonium-239** for Stage 2. • operational through its fleet of **Pressurised Heavy Water Reactors**.
Stage 2	**Fast Breeder Reactors (FBRs)** using **Plutonium-239**	• Plutonium from Stage 1 is used as fuel. • A blanket of Uranium-238 surrounds the core and gets converted into more Plutonium-239.	• Breed more fuel than is consumed and prepare **Uranium-233** for Stage 3. • **500 MWe Prototype Fast Breeder Reactor (PFBR) at Kalpakkam attained first criticality on 6 April 2026**,

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		• The reactor is designed to eventually use Thorium-232 in the blanket. • Through **transmutation, Thorium-232 will be converted into Uranium-233,** the fuel that will power India's third stage of nuclear energy based on thorium.	marking a key step toward the breeder stage.
Stage 3	**Thorium-based reactors** using **Uranium-233 and Thorium-232**	• Thorium-232 is not fissile by itself, but it can be converted into fissile Uranium-233. • This Uranium-233 then sustains the thorium fuel cycle.	• Achieve large-scale use of India's abundant thorium reserves. • **Advanced Heavy Water Reactor (AHWR)** and **Indian Molten Salt Breeder Reactor (IMSBR)** technologies are being developed to demonstrate and eventually scale thorium-based power generation.

A New Chapter in India's Nuclear Journey

(Indigenously built 500 MWe fast breeder reactor attains first criticality)

What is criticality:

- **Criticality** means the initiation of a **self-sustaining nuclear fission reaction** that will eventually lead to the generation of power by the **500 MWe FBR**.
    - It indicates that the reactor core is functioning as designed and that each fission event now releases sufficient neutrons to sustain an ongoing series of reactions.

Significance:

- With this achievement, India has officially entered the second stage of its three-stage nuclear power programme.
- Once fully operational, India will become **only the second country in the world after Russia** to operate a commercial fast breeder reactor.
- This achievement is a testament to the decades of scientific effort led by the Department of Atomic Energy.
- It also marks a significant step in India's clean energy journey, reinforcing the country's commitment to reliable, low-carbon power.
- Additionally, it brings India closer to its goal of **achieving net zero emissions by 2070.**

Key Challenges

Challenge	Explanation
Thorium is fertile, not fissile	• Thorium-232 cannot directly sustain a chain reaction. It must first be converted into Uranium-233.
Complex breeder technology	• Fast breeder reactors involve advanced fuel design, sodium coolant technology, reprocessing, and high safety requirements. • Countries such as **Japan, France, and the United States** had shut down fast breeder programmes amid safety concerns.
Slow transition	• Large-scale thorium use depends on the successful scaling of Stage 2 because Stage 3 needs adequate Uranium-233.

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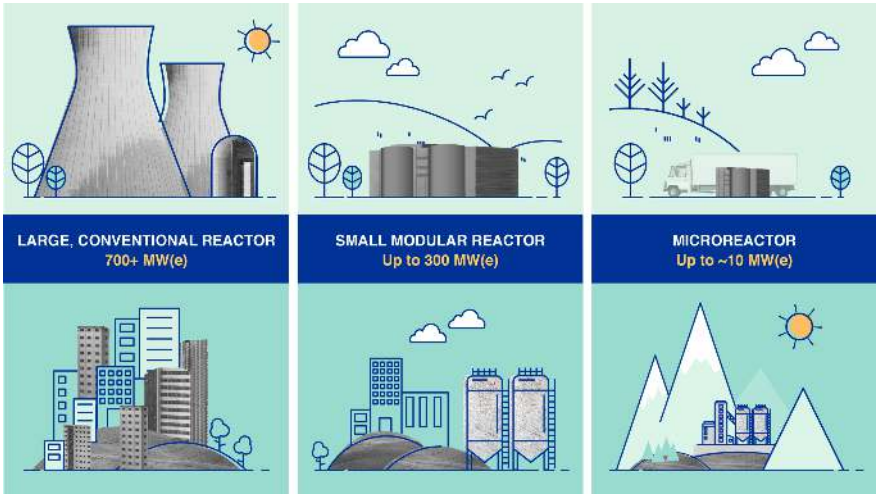
Cost and safety concerns	• Nuclear expansion requires strict regulatory oversight, high capital investment, and public confidence.

Conclusion

India's three-stage nuclear programme is a **resource-smart, self-reliant and long-term energy strategy**. Its success can convert India's thorium advantage into clean baseload power, but it depends on timely scaling of fast breeder reactors, safe reprocessing and commercialisation of thorium-based technologies.

Small modular reactors (SMRs)

- Recent Context: India's 2025 Budget launched the **Nuclear Energy Mission** with a ₹20,000 crore outlay to achieve **100 GW nuclear capacity by 2047** and deploy **five indigenously manufactured SMRs by 2033**.
- SMRs are a**dvanced nuclear reactors** that have a power capacity of up to 300 MW(e) per unit.
- According to the International Atomic Energy Agency (IAEA), more than 70 SMR concepts are currently under development in 18 countries.
- **Advantages of SMRs**
  - **Smaller footprint:** SMRs can be sited on locations not suitable for larger nuclear power plants.
  - **Affordable:** Prefabricated units of SMRs can be manufactured and then shipped and installed on site.
- **Savings**: SMRs offer savings in **cost** and **construction time**



Bharat Small Modular Reactor

- The **Bharat Small Modular Reactor (BSMR)** is a new generation of compact nuclear reactors designed and developed indigenously by India, primarily through the **Bhabha Atomic Research Centre (BARC)** and **Nuclear Power Corporation of India Limited (NPCIL).**
- The flagship model, BSMR-200, is a 200 MWe Pressurised Heavy Water Reactor (PWR) that uses slightly enriched uranium fuel.
- In addition to the 200 MWe BSMR-200, a 55 MWe variant is being developed for smaller, remote applications.
- **Deployment**: The lead unit will be at a **Department of Atomic Energy (DAE)** site. Subsequent units may be sited at industrial facilities or repurposed thermal plant locations.

Core Technology and Design

- **Type:** 220 MW Pressurized Heavy Water Reactor (PHWR)
- **Proven Track Record:** Based on India's established PHWR technology.
- **Upgrades:** The design is being refined to reduce land requirements, making BSRs suitable for deployment near industrial hubs.

Advantages of BSRs

- **Faster Deployment:** Smaller size and standardized design allow for quicker construction and commissioning

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compared to traditional large nuclear plants.

- **Passive Safety:** BSMR includes systems that can maintain safety functions without operator intervention or external power, enhancing resilience.
- **Cost-Effectiveness:** Lower capital requirements and an established domestic supply chain make BSRs potentially more affordable.
- **Grid Flexibility:** BSRs are well-suited for both grid-connected and off-grid applications, enhancing energy access in remote or underserved regions.
- **Decarbonization:** Provides a reliable, low-carbon alternative to fossil fuels for baseload power, crucial for India's climate goals.
- **Waste Management:** In-situ storage and reprocessing capabilities are planned to minimize waste and recover useful radioisotopes.

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Quantum Technology

Meaning

- **Quantum Technology** refers to technologies that use the principles of **quantum mechanics** to process information, communicate securely, measure precisely and design advanced materials.
- It works at the level of **atoms, photons, electrons, ions and molecules**.
- It is different from classical technology because it uses unusual quantum behaviour such as **superposition, entanglement and tunnelling**.

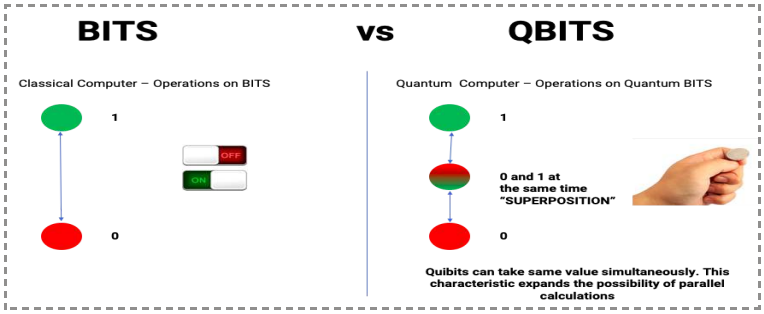
Basics of Quantum Technology

Quantum

- A **quantum** is the smallest discrete unit of energy or matter behaviour.
- Example: A **photon** is the quantum of light.
- Quantum technology controls such tiny particles to perform tasks that are difficult for classical systems.

Qubit

- A **qubit** is the basic unit of quantum information.
- A classical bit can be either **0 or 1**.
- A qubit can exist in a combination of **0 and 1 at the same time**.
- This gives quantum computers the potential to process complex possibilities faster.



Superposition

- Superposition means a quantum particle can exist in multiple states at the same time until it is measured.
- It allows quantum computers to explore many possible solutions simultaneously.
    - **Example:** Useful in optimisation problems, drug discovery and artificial intelligence.

Entanglement

- Entanglement means two quantum particles become connected in such a way that the state of one affects the other, even if they are far apart.
- It is the foundation of **quantum communication** and **Quantum Key Distribution**.

Quantum Key Distribution (QKD)

- It is a method of secure communication based on quantum principles.
- It allows two parties to create and share a secret encryption key.
- Its security comes from the fact that any attempt to observe or measure a quantum state disturbs it.
- Therefore, if a hacker tries to intercept the key, the sender and receiver can detect the intrusion.
- QKD does not transmit the actual message; it only helps in securely sharing the encryption key.
- It is useful for defence communication, banking networks, diplomatic communication, space communication and critical infrastructure.
- **IIT-Delhi and DRDO** successfully demonstrated **Quantum Key Distribution (QKD)** over **1 km of free space**, a first-of-its-kind achievement in India.

Interference

- Interference means quantum states can combine with each other.
- Correct solutions can be amplified, while wrong solutions can be cancelled.
- It is useful in quantum algorithms.

Tunnelling

- Tunnelling means a quantum particle can pass through an energy barrier that it should not cross according to classical physics.
- It is useful in quantum devices, sensors, and advanced electronics.

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Decoherence

- Decoherence means loss of quantum behaviour due to interaction with the external environment.
- It is one of the biggest challenges in making stable quantum computers.
- Quantum systems require isolation, extremely low temperature, and error correction.

Achievements and Initiatives by India

National Quantum Mission

- The **National Quantum Mission** is India's flagship programme for quantum technology.
- Approved with an outlay of **₹6003.65 crore** for eight years.
- It aims to build India's capacity in:
    - Quantum computing.
    - Quantum communication.
    - Quantum sensing and metrology.
    - Quantum materials and devices.
- It seeks to create a complete ecosystem of research, start-ups, skilled manpower, testbeds and industry participation.
- **Four Thematic Hubs:**
    - Quantum computing.
    - Quantum communication.
    - Quantum sensing and metrology.
    - Quantum materials and devices.
- **Quantum Computing Targets: India aims to develop quantum computers with:**
    - **20–50 physical qubits** in the initial phase.
    - **50–100 physical qubits** in the medium term.
    - **50–1000 physical qubits** in the long term.
- This will help India move from research dependence to indigenous quantum hardware development.

Quantum Communication Targets

- India aims to develop satellite-based secure quantum communication over long distances.
- It also targets inter-city Quantum Key Distribution networks.
- This is important for secure defence, government and banking communication.

Quantum Communication Demonstrations

- India has demonstrated progress in fibre-based and free-space quantum communication.
- India has also demonstrated inter-city and long-distance quantum communication links.

QSim Quantum Simulator

- India launched the **QSim Quantum Computer Simulator Toolkit**.
- It helps researchers and students write, test and debug quantum algorithms.
- It is useful because access to real quantum computers is limited and expensive.

Quantum Diamond Microscope

- India has developed indigenous capability in advanced quantum sensing tools like the Quantum Diamond Microscope.
- It can be useful in magnetic-field imaging, material science, biology and semiconductor research.

Significance for India

- Quantum technology reduces **dependence on foreign-controlled digital infrastructure.**
- It strengthens India's **technological sovereignty in computing, cybersecurity and communication.**
- Future quantum computers may break present encryption methods like **Rivest-Shamir-Adleman encryption.**
    - India must prepare for **Harvest Now, Decrypt Later** attacks, where adversaries store encrypted data today and decrypt it later using quantum computers.
- Secure communication, quantum radar, quantum navigation, and quantum sensors can improve defence capability.
- It is important for **border security, submarine detection, missile systems, and command networks.**
- Quantum-safe communication can protect government databases, financial systems and critical digital infrastructure.

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- It supports India's goal of a secure and **trusted digital public infrastructure.**
- Quantum technology can create demand for **lasers, cryogenic systems, photonics, semiconductors and advanced materials.**

Challenges

- **Fragile quantum systems:** Qubits are easily disturbed by heat, noise and external interference, leading to decoherence and errors.
- **Hardware dependence:** India needs stronger indigenous capacity in cryogenics, photonics, lasers, superconducting circuits, semiconductor fabrication and precision electronics.
- **Skilled manpower gap:** Quantum technology requires interdisciplinary expertise in physics, computer science, electronics, material science and engineering.
- **High cost and long gestation:** Quantum labs, clean rooms, low-temperature systems and precision instruments require heavy investment with uncertain commercial returns.
- **Cybersecurity transition risk:** Future quantum computers may break existing encryption, creating risks for defence, banking, telecom and digital public infrastructure.
- **Weak industry-academia linkage:** Research may remain confined to laboratories unless connected with start-ups, defence, telecom, finance and manufacturing sectors.

Way Forward

- **Implement National Quantum Mission in mission mode** with clear timelines, measurable outcomes, periodic review and inter-ministerial coordination.
- **Build indigenous hardware ecosystem** in cryogenics, photonics, lasers, semiconductors, quantum materials and precision electronics.
- **Start quantum-safe cybersecurity transition** through national audit of vulnerable encryption systems and phased adoption of post-quantum cryptography.
- **Develop skilled human resources** through interdisciplinary courses, fellowships, quantum labs and industry-oriented training.
- **Promote start-ups and industry participation** through challenge grants, public procurement, testbeds, sandboxes and defence-sector use cases.
- **Prioritise near-term applications** such as Quantum Key Distribution, quantum sensors, quantum-safe encryption and quantum simulation instead of focusing only on universal quantum computers.

Conclusion

Quantum technology is a frontier technology with major implications for cybersecurity, defence, communication, healthcare, finance and strategic autonomy. For India, the priority should be to move from **scientific research to technological capability and commercial deployment**. Effective implementation of the National Quantum Mission can help India become a major player in the emerging global quantum order.

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